# Project Draft

## Data Overview

```r
# packages being used
library(tidyverse)
library(corrplot)
library(dplyr)
library(GGally)
library(car)

NBA_Player_Stats <- read.csv("Datasets/2025-2026 NBA Player Stats - NBAstuffer.csv")
NBA_Player_Stats <- NBA_Player_Stats[, -1]
NBA_Player_Contracts <- read.csv("Datasets/2025-26 NBA Player Contracts.csv")
result <- merge(NBA_Player_Stats, NBA_Player_Contracts, by.x = "NAME", by.y = "Player")
head(result)
```

```
          NAME TEAM CUR POS  AGE GP  MpG USG.  TO. FTA   FT. X2PA  X2P. X3PA
1    A.J. Green  Mil   *   G 26.5 60 29.5 13.3  9.7  41 0.805   46 0.500  418
2  Aaron Gordon  Den   *   F 30.5 26 27.5 22.9  7.8 121 0.793  183 0.557  114
3 Aaron Holiday  Hou   *   G 29.4 43 13.0 17.9 11.2  37 0.865   72 0.458  125
4 Aaron Nesmith  Ind   * G-F 26.4 37 29.6 19.7 10.5  97 0.825  192 0.427  227
5 Aaron Wiggins  Okc   *   G 27.2 51 23.5 19.8 12.7  66 0.742  223 0.502  231
6    Ace Bailey  Uta   *   F 19.6 58 27.1 20.1 10.6  62 0.742  375 0.517  282
   X3P.  eFG.   TS.  PpG RpG ApG SpG BpG TOpG  P.R  P.A P.R.A  VI  ORtg  DRtg
1 0.407 0.599 0.611  9.8 2.5 1.9 0.5 0.1  0.8 12.3 11.7  14.2 4.9 117.0 116.3
2 0.386 0.566 0.617 16.6 6.1 2.5 0.7 0.2  1.0 22.7 19.1  25.2 9.2 128.1 113.8
3 0.360 0.510 0.546  5.4 0.9 0.9 0.5 0.1  0.6  6.3  6.3   7.2 5.0 114.5 108.7
4 0.352 0.482 0.524 13.1 4.5 2.1 0.6 0.6  1.3 17.5 15.2  19.6 6.7 106.0 114.0
5 0.355 0.518 0.537 10.2 3.4 1.8 0.9 0.5  1.3 13.5 12.0  15.4 6.8 104.6 106.4
6 0.344 0.517 0.530 12.5 4.0 1.7 0.9 0.5  1.3 16.5 14.2  18.2 6.5 106.1 116.9
  Rank Team  X2025.26  X2026.27  X2027.28  X2028.29  X2029.30 X2030.31
1  355  MIL  $2301587 $10044644 $10848215 $11651786 $12455355
2   82  DEN $22841455 $33658037 $36350680 $39043323
3  386  HOU  $2296274
4  158  IND $11000000 $11000000 $19418000 $20971440
5  172  OKC $10102803  $9224300  $8345797  $8345797
6  184  UTA  $9069840  $9523080  $9976560 $12640302
  Guaranteed Additional
1  $47301587  greenaj01
2  $92850172  gordoaa01
3   $2296274  holidaa01
4  $62389440  nesmiaa01
5  $27672900  wiggiaa01
6  $18592920  baileac01
```

```r
dim(result)
```

```
[1] 547  39
```

We decided to do Bayesian Inference on this question: Can we use the posterior predictive distribution to identify players who are significantly overpaid or underpaid relative to their market value?

First, we merged the two datasets we found into one dataset. Then we perform EDA on the dataset, explore the distribution of variables to find if there exists patterns, outliers, and we would also want to find if there's any relationship between variables which would influence our later on analysis. Data cleaning is also conducted at this step as well by dropping some columns that are unrelated to our research question and removing the outliers we find.

The two methods we decided to use are:

1. Using stan to build a Bayesian linear regression model.

   Before building the model, we will do variable selection based off the definition of the variables, and using methods like VIFs, forward model selection to choose the appropriate variables for the model. Then we will use the model to do stan,

choosing the appropriate prior and posterior for the variables, and do posterior predictive distribution to identify players who are significantly overpaid or underpaid relative to their market value.

2. MCMC Metropolis-Hastings

   We decided to do something related to MH algorithm. We implemented the Metropolis-Hastings algorithm manually to sample from the posterior distribution. Then, compare the MCMC result with the Stan method result. This allows us to determine whether or not those two methods produce consistent posterior estimates.

## Data Cleaning

We notice that our contracted salary columns (X2025.26 to X2030.31) are treated as character instead of numeric because of the extra dollar sign ($). We must remove the sign and set these to numeric for later analysis.

```
result <- result %>%
  mutate(across(starts_with("X20"), ~ as.numeric(gsub("\\$", "", .x))))
class(result$X2025.26)
```

```
[1] "numeric"
```

We also need to check whether there are any NA values for our response variable; if NA exists we would need to drop that row.

```
result[is.na(result$X2025.26), ]
```

```
               NAME TEAM CUR POS  AGE GP MpG USG. TO. FTA FT. X2PA X2P. X3PA
88 Chris Livingston  Cle   *   F 22.4  3 5.8 17.9   0   1   1    4    1    3
   X3P.  eFG.   TS. PpG RpG ApG SpG BpG TOpG P.R P.A P.R.A  VI ORtg DRtg Rank
88    0 0.571 0.605   3   1 0.3 0.3   0    0   4 3.3   4.3 6.7  129   91  381
   Team X2025.26 X2026.27 X2027.28 X2028.29 X2029.30 X2030.31 Guaranteed
88  MIL       NA       NA       NA       NA       NA       NA   $2296274
   Additional
88  livinch01
```

There exists one row which misses the value for our response variable, we drop this observation.

```
result_final <- result[!is.na(result$X2025.26), ]
dim(result_final)
```

```
[1] 546  39
```

Next, we select the relevant columns for our research question and rename them for clarity.

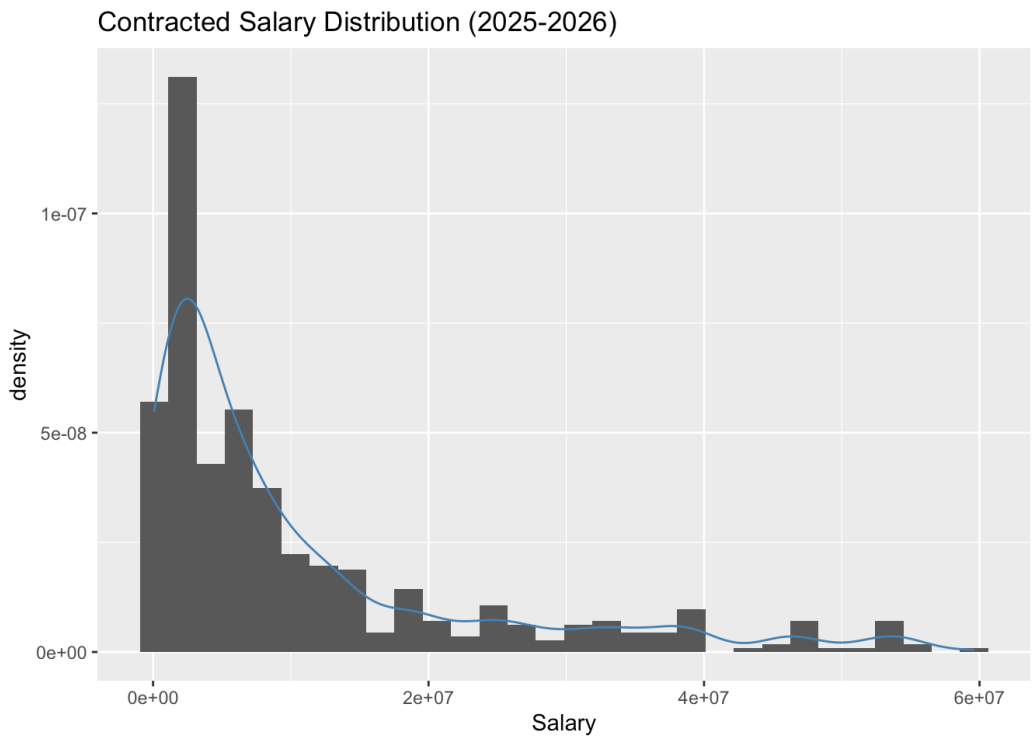
```
NBA_Player_Stats_clean <- result_final[, c("NAME", "TEAM", "CUR", "POS", "AGE", "GP", "MpG", "USG.", "TO.", "T

df <- NBA_Player_Stats_clean
df <- rename(df, Salary = X2025.26)   # rename salary
names(df) <- sub("\\..*", "", names(df))  # remove dots in names
head(df)
```

```
           NAME TEAM CUR POS  AGE GP  MpG  USG   TO    TS  PpG RpG ApG SpG BpG
1    A.J. Green  Mil   *   G 26.5 60 29.5 13.3  9.7 0.611  9.8 2.5 1.9 0.5 0.1
2  Aaron Gordon  Den   *   F 30.5 26 27.5 22.9  7.8 0.617 16.6 6.1 2.5 0.7 0.2
3 Aaron Holiday  Hou   *   G 29.4 43 13.0 17.9 11.2 0.546  5.4 0.9 0.9 0.5 0.1
4 Aaron Nesmith  Ind   * G-F 26.4 37 29.6 19.7 10.5 0.524 13.1 4.5 2.1 0.6 0.6
5 Aaron Wiggins  Okc   *   G 27.2 51 23.5 19.8 12.7 0.537 10.2 3.4 1.8 0.9 0.5
6    Ace Bailey  Uta   *   F 19.6 58 27.1 20.1 10.6 0.530 12.5 4.0 1.7 0.9 0.5
    Salary
1  2301587
2 22841455
3  2296274
4 11000000
5 10102803
6  9069840
```

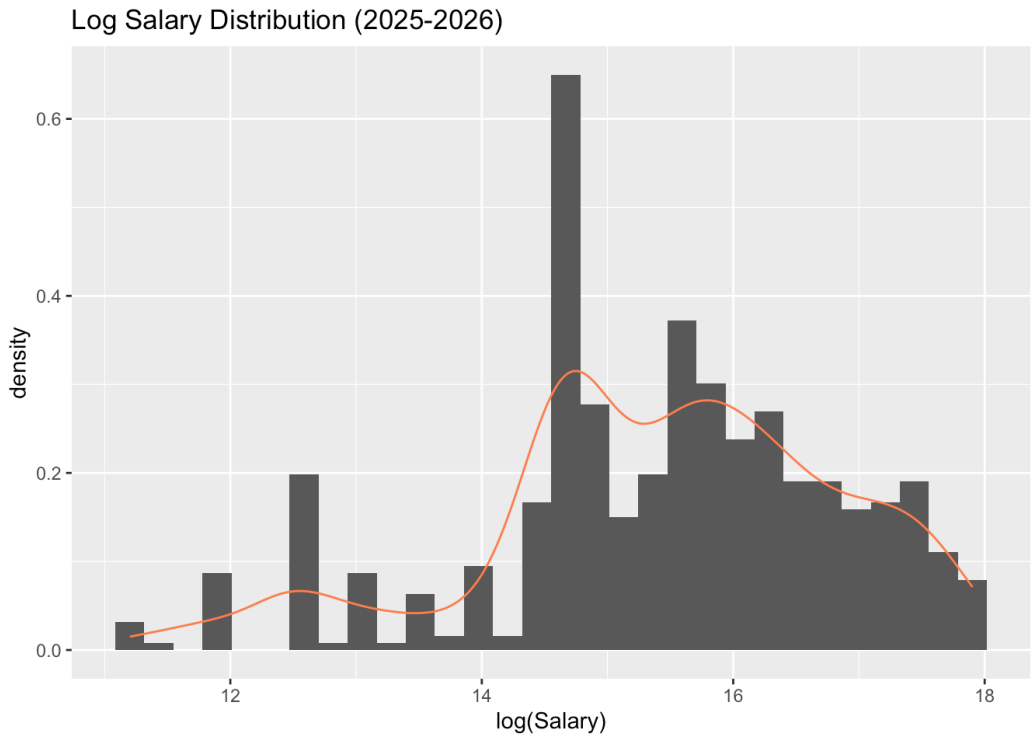
# EDA

## Salary Distribution

```
# Original Salary
p1 <- ggplot(df, aes(x = Salary)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30) +
  geom_density(color = "steelblue") +
  labs(title = "Contracted Salary Distribution (2025-2026)")

p1
```



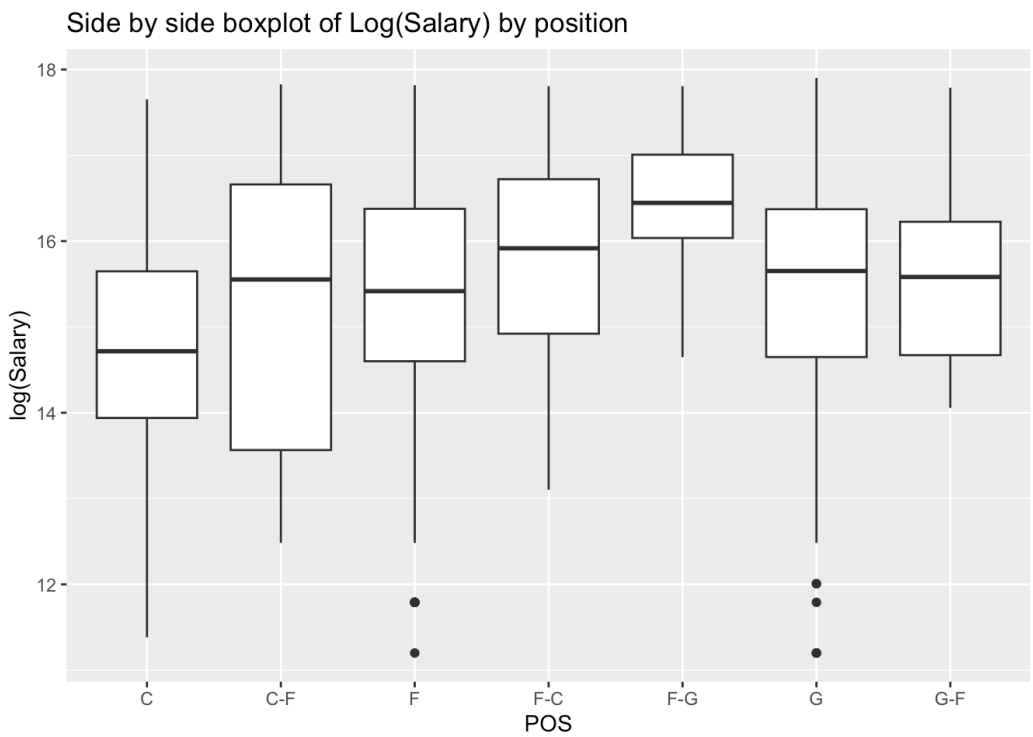

```
# Log Salary
p2 <- ggplot(df, aes(x = log(Salary))) +
  geom_histogram(aes(y = after_stat(density)), bins = 30) +
  geom_density(color = "coral") +
  labs(title = "Log Salary Distribution (2025-2026)")

p2
```




We notice that the distribution of the contracted salary (X2025.26) is heavily right-skewed. A log transformation is necessary in order to standardize the variable.

## Salary by Position

```
# Position
ggplot(df, aes(x = POS, y = log(Salary))) +
  geom_boxplot() +
  labs(title = "Side by side boxplot of Log(Salary) by position")
```



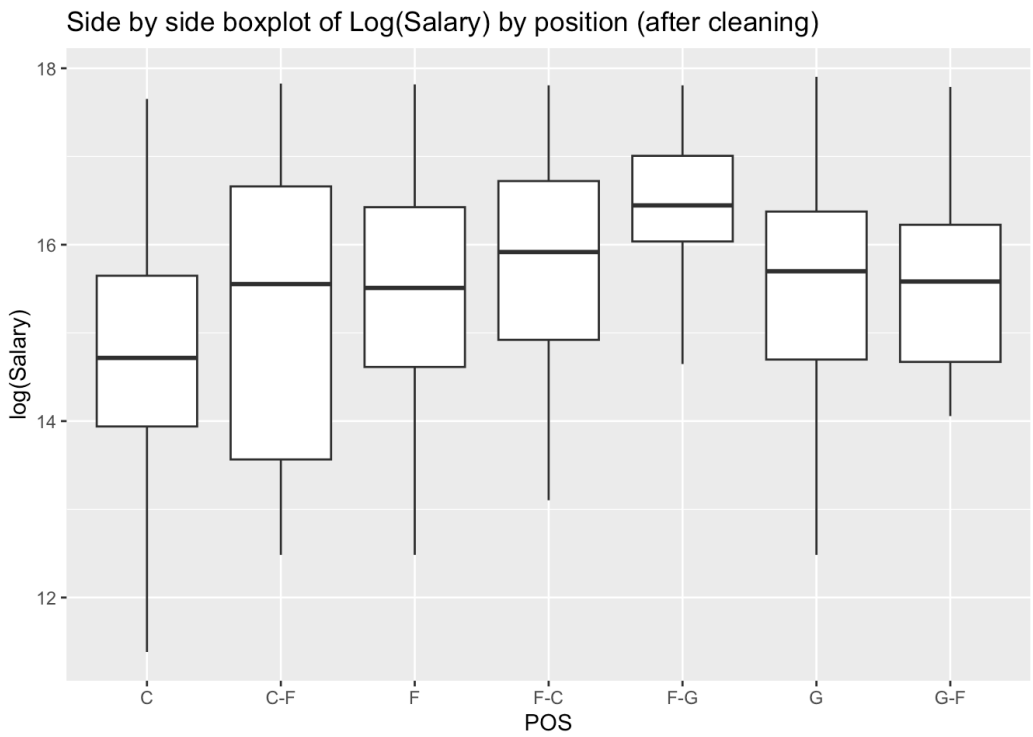

## Outliers Removal

```
# remove outliers
df <- df %>%
```

```
  group_by(POS) %>%
  mutate(
    Q1 = quantile(log(Salary), 0.25, na.rm = TRUE),
    Q3 = quantile(log(Salary), 0.75, na.rm = TRUE),
    IQR = Q3 - Q1,
    lower = Q1 - 1.5 * IQR
  ) %>%
  filter(!(POS %in% c("F", "G") & log(Salary) < lower)) %>%
  select(-Q1, -Q3, -IQR, -lower) %>%
  ungroup()
dim(df)
```

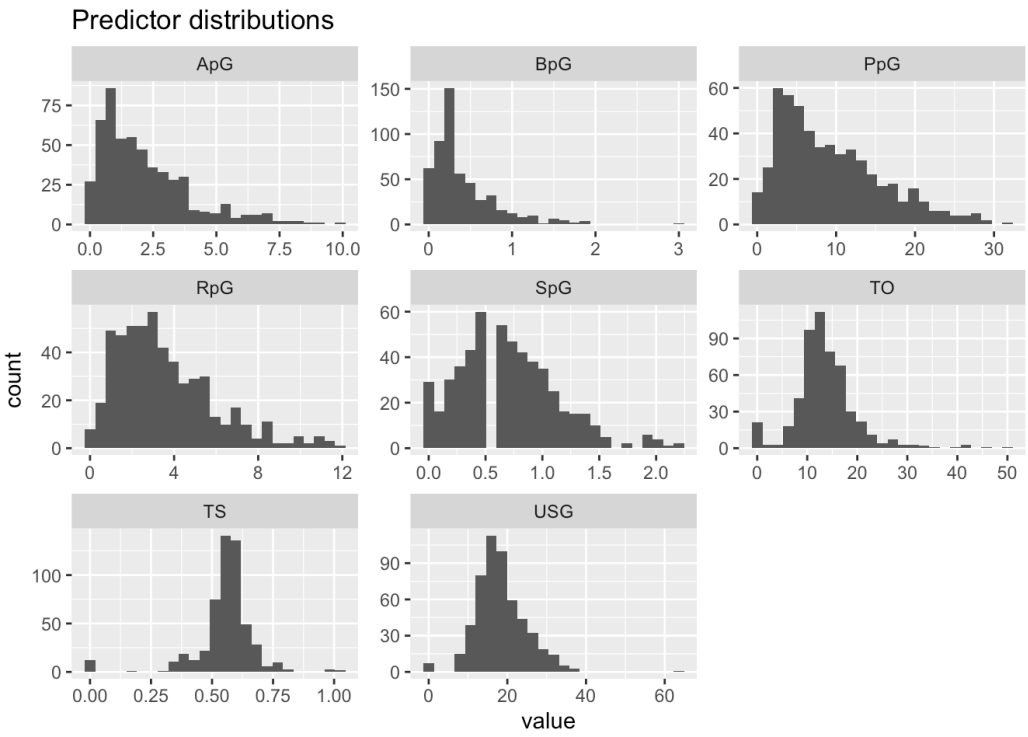
```
[1] 531  16
```

```
# Position boxplot after data cleaning
ggplot(df, aes(x = POS, y = log(Salary))) +
  geom_boxplot() +
  labs(title = "Side by side boxplot of Log(Salary) by position (after cleaning)")
```

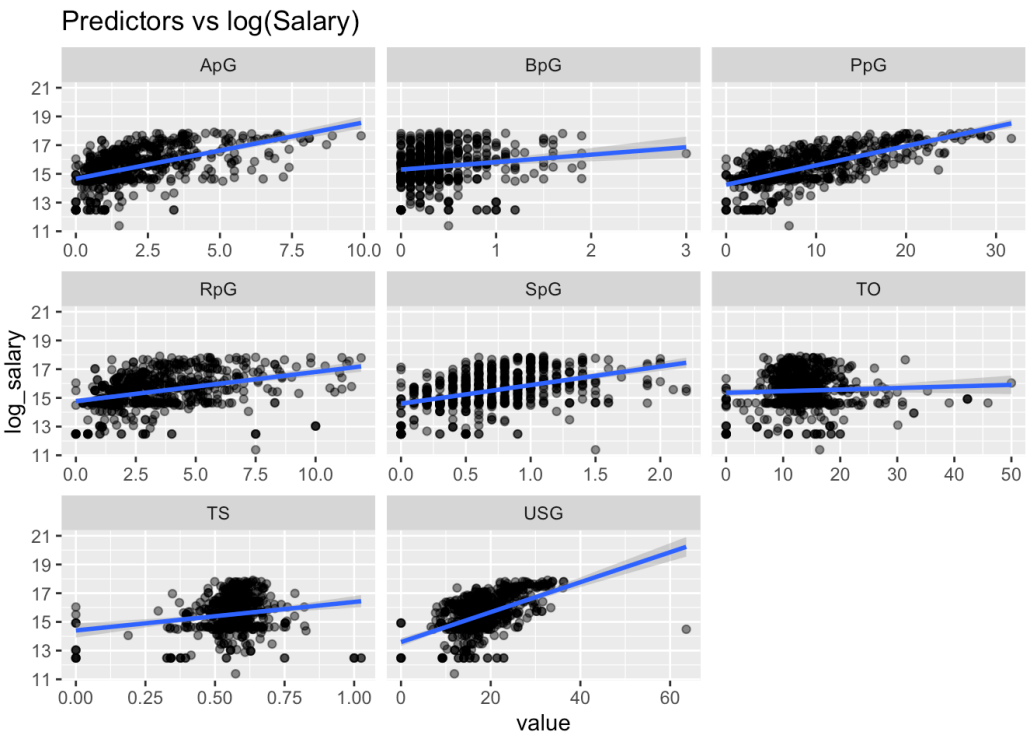



## Predictor Distributions and Relationships with Log(Salary)

```
pred_vars <- c("USG", "TO", "TS", "PpG", "RpG", "ApG", "SpG", "BpG")

# Histogram
df %>%
  select(all_of(pred_vars)) %>%
  pivot_longer(everything()) %>%
  ggplot(aes(x = value)) +
  geom_histogram(bins = 25) +
  facet_wrap(~name, scales = "free") +
  labs(title = "Predictor distributions")
```

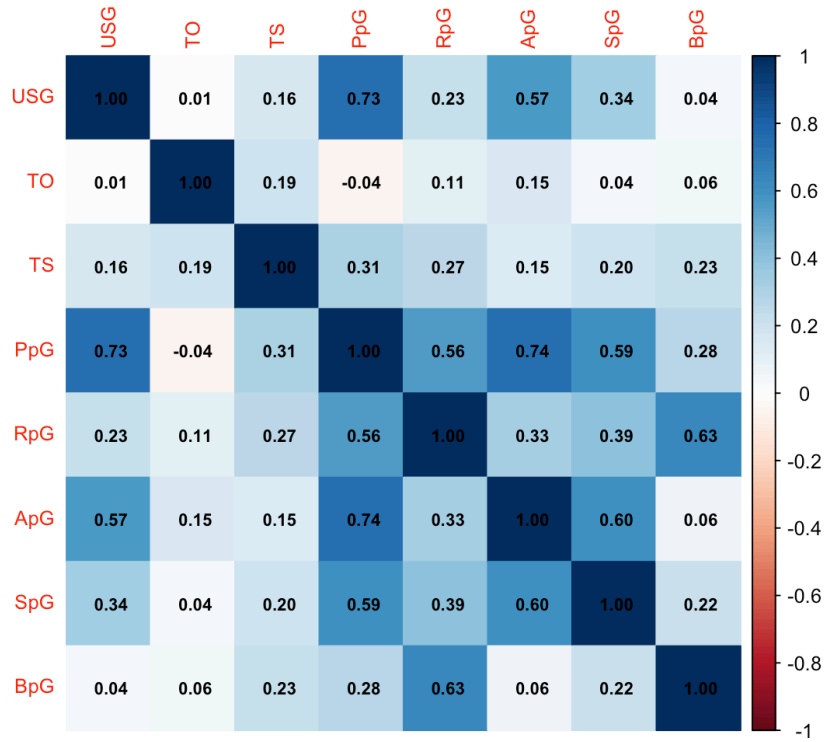



```
# Variables vs log(Salary) Scatterplot
df %>%
  mutate(log_salary = log(Salary)) %>%
  select(log_salary, all_of(pred_vars)) %>%
  pivot_longer(-log_salary) %>%
  ggplot(aes(x = value, y = log_salary)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = TRUE) +
  facet_wrap(~name, scales = "free_x") +
  labs(title = "Predictors vs log(Salary)")
```

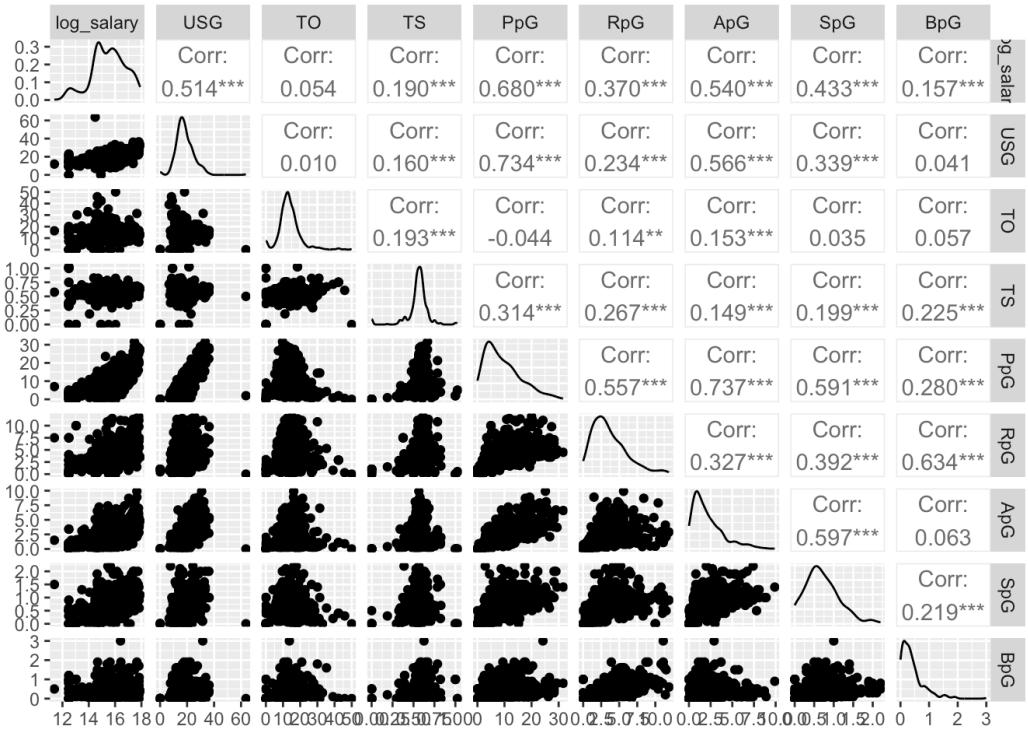



## Correlation and Multicollinearity Check

```r
# Correlation Plot / Matrix
cor_mat <- cor(df[, pred_vars], use = "complete.obs")
corrplot::corrplot(cor_mat, method = "color", addCoef.col = "black",
                   number.cex = 0.7, tl.cex = 0.8)
```




```r
# Pairs plot (log Salary)
df %>%
  mutate(log_salary = log(Salary)) %>%
  select(log_salary, all_of(pred_vars)) %>%
  ggpairs()
```




```r
# VIF
ols_fit <- lm(log(Salary) ~ USG + TO + TS + PpG + RpG + ApG + SpG + BpG,
```

```
              data = df)
vif(ols_fit)
```

```
     USG       TO       TS      PpG      RpG      ApG      SpG      BpG
2.543840 1.244512 1.239870 6.016229 2.464104 2.920397 1.777892 1.773535
```

We notice that for the heat map, there's no correlation coefficient greater than $0.8$, which means that all the variables at this stage are not considered to be highly correlated to each other. So we decide not to drop any of them.

Note that for our VIF values, there exists one VIF value of $6.016229$ (`PpG`) which is normally considered to be a concerning threshold that requires further investigation since this variable could potentially be affecting coefficient estimates. Since `PpG` represents points per game, higher `PpG` surely results in higher `Salary`. However, we decide not to drop this column, since this VIF value is not extremely high (less than 10), and this variable acts as a key factor to consider for our research question.

## Model 1: Bayesian Linear Regression

Our naive model is linear regression.

$$\log(salary)_i \sim \mathcal{N}(\alpha + \beta_1 \cdot USG_i + \beta_2 \cdot TO_i + \beta_3 \cdot TS_i + \beta_4 \cdot PpG_i + \beta_5 \cdot RpG_i + \beta_6 \cdot ApG_i + \beta_7 \cdot SpG_i + \beta_8 \cdot BpG_i,\ \sigma^2)$$

And we have

$$\alpha \sim \mathcal{N}(16, 2)\beta_1 \sim \mathcal{N}(), \beta_2 \sim \mathcal{N}(), \beta_3 \sim \mathcal{N}(), \beta_4 \sim \mathcal{N}(), \beta_5 \sim \mathcal{N}(), \beta_6 \sim \mathcal{N}(), \beta_7 \sim \mathcal{N}(), \beta_8 \sim \mathcal{N}(), \sigma \sim ()$$

## Model 2: Hierarchical Bayesian Linear Regression

Our complex model is Hierarchical Linear Regression with position-specific intercepts.

$$\log(salary)_i \sim \mathcal{N}(\alpha_j + \beta_1 \cdot USG_i + \beta_2 \cdot TO_i + \beta_3 \cdot TS_i + \beta_4 \cdot PpG_i + \beta_5 \cdot RpG_i + \beta_6 \cdot ApG_i + \beta_7 \cdot SpG_i + \beta_8 \cdot BpG_i,\ \sigma^2)$$

And we have

$$\alpha_j \sim \mathcal{N}(\mu_a, \tau^2)\beta_1 \sim \mathcal{N}(), \beta_2 \sim \mathcal{N}(), \beta_3 \sim \mathcal{N}(), \beta_4 \sim \mathcal{N}(), \beta_5 \sim \mathcal{N}(), \beta_6 \sim \mathcal{N}(), \beta_7 \sim \mathcal{N}(), \beta_8 \sim \mathcal{N}(), \tau \sim ()$$